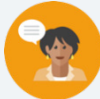


9.4 Solids of Rotation

Lesson Introduction

 Benchmark	MA.912.GR.4.2 Identify three-dimensional objects generated by rotations of two-dimensional figures.		 Learning Target	I can identify three-dimensional objects generated by rotations of two-dimensional figures.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How can you identify three-dimensional objects generated by rotations of two-dimensional figures? - How can you identify the two-dimensional figure that was used to generate a three-dimensional object by rotation?			
 Lesson Narrative	This lesson allows students to use cross sections and has students perform rotations to see what three-dimensional objects would be formed by a rotation around a given axis. Students discover the three-dimensional objects formed by various two-dimensional figures. The lesson also has students work backward by giving three-dimensional objects, and students have to determine what two-dimensional figure would have formed the three-dimensional object by rotation.			
 Component Summary	9.4.1 Warm-Up (~5 min.)	Sketching cross sections of a Rubik’s Cube and a Slinky (<i>SE p. 71</i>)	9.4.3 Guided Instruction (10 min.)	Sketching two-dimensional figures or three-dimensional objects that create or result from rotations (<i>SE p. 74</i>)
	9.4.2 Activity (15-20 min.)	Describing and sketching three-dimensional objects created by rotating two-dimensional figures (<i>SE p. 71</i>)	9.4.4 Guided Practice (5-10 min.)	Describing and sketching three-dimensional objects created by rotating two-dimensional figures (<i>SE p. 75</i>)
	9.4.5 Wrap-Up (~5 min.)	Describing and sketching three-dimensional objects created by rotating two-dimensional figures (<i>SE p. 75</i>)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		- Sticky Notes - Pencils (Optional) Graphing Utility (Optional) Honeycomb Paper Decorations (Optional) Drill and Cardboard Rectangle, Triangle, and Semicircle	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may mistake the vertical axis and the horizontal axis. - Students may incorrectly graph two-dimensional figures and three-dimensional objects due to not using the correct dimensions, such as height and radius.	Consider using honeycomb paper decorations to model solids of rotation.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder This lesson provides a plethora of opportunities for students to notice and wonder, especially in 9.4.1 Warm-Up and 9.4.2 Activity. The lesson opens with a Notice and Wonder, and although not specifically stated in the 9.4.2 Activity, students are required to make predictions, which requires students to notice and wonder.	MA.K12.MTR.4.1: Discussion and Reflection This lesson is discussion-heavy, especially throughout 9.4.2 Activity. Students are required to determine which three-dimensional object would be formed through a rotation and how that would be sketched. Students will also analyze the mathematical thinking of other students throughout this lesson.
 Differentiate Instruction	Consider providing visuals and other manipulatives to create kinesthetic entry-points for students. Using visuals can help all students better visualize how solids of rotation work. For example, digital applets like GeoGebra’s Solids of Rotation offer students the chance to digitally explore the rotation two-dimensional figures and the three-dimensional objects that they form.	
Lesson Notes:		

9.4.1 Warm-Up: Notice and Wonder

Component Overview: Students complete a Notice and Wonder given a picture of a Rubik's Cube and Slinky. Students are also expected to sketch the horizontal cross section of the cube and the Slinky.



9.4.1 Warm-Up: Notice and Wonder

A Rubik's Cube and a Slinky are shown.

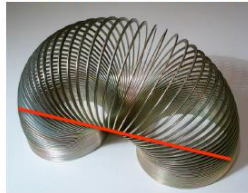


1. What do you notice?
Answers will vary. They both could form patterns.
2. What do you wonder?
Answers will vary. How could they be turned or stretch to make new patterns?
3. Sketch the horizontal cross section of the Rubik's cube.



The red lines are on picture will form a square.

4. Sketch the horizontal cross section of the Slinky.



Answer will vary.



Allow students to share their notices and wonders from questions 1 and 2 prior to sketching the cross sections in questions 3 and 4.



For question 4, students may interpret the cross section differently. Some may see the cross section as the shape of an "athletic track" while others may see an oval. If time permits, entertain a quick discussion of what they see.



Consider asking the following questions:

- What would the horizontal cross section of the Slinky be if it were in a stacked position and not stretched over? (Circle.)
- What would the vertical cross section of the Slinky be if it were in a stacked position and not stretched over? (Rectangle.)

9.4.2 Activity: Rotating Pencils

Component Overview: 9.4.2 Activity has students working with partners to discover solids of rotations by rotating pencils. Students will be given sticky notes (or a 3x5 index card) to tape to the side of a pencil. Students will answer guided questions to discover which three-dimensional objects will be formed with rotations.



9.4.2 Activity: Rotating Pencils

Work with your partner to complete the following.

1. Place a sticky note on the side of the pencil. If necessary, use tape to secure the sticky note.
2. Make a prediction about which three-dimensional shape the sticky note would form when the pencil is rotated 360° .

Complete the table.

	Prediction	Reasoning
Sticky Note	<i>Answers will vary. cylinder</i>	<i>Answers will vary. Going around the pencil makes a circle. With it being a pencil to hold and a 3-dimensional shape, it is in the form of a cylinder.</i>

3. Using your pencil as a vertical axis of rotation, rotate the pencil 360° .
4. What three-dimensional figure did the sticky note seem to create?
cylinder



Throughout 9.4.2 Activity, students will complete questions by rotating their pencils and noticing and wondering to make their predictions and observations.

9.4.2 Activity: Rotating Pencils (continued)

5. Work with your partner to predict what three-dimensional shape will be formed when each shape is rotated 360° .

Shape	Prediction	Reasoning
Rectangle	<i>Cylinder</i>	<i>Answers will vary. As a rectangle rotates around it forms a circular shape of a cylinder.</i>
Triangle	<i>Cone</i>	<i>Answers will vary. When it is rotated, it forms a circular base with triangular sides.</i>
Semicircle	<i>Sphere</i>	<i>Answers will vary. Half of a sphere is a hemisphere since a semicircle is half of a circle, but when it rotates a sphere will form.</i>



Consider opportunities to connect student predictions to visual representations such as the honeycomb paper decorations or applets, like GeoGebra's Solids of Rotations.



Consider allowing students to share their responses before having students rotate their pencils to see if students' predictions match the resulting three-dimensional objects.

6. Using your pencil as a vertical axis of rotation, rotate the pencil 360° . Then, sketch or describe the resulting three-dimensional shape.

Shape	Three-Dimensional Shape
Rectangle	<i>Cylinder. It's like folding a rectangle around a pencil, and it taking the shape of the pencil.</i>
Triangle	<i>Cone. As the triangle rotates around the pencil, it's like forming an ice cream cone.</i>
Semicircle	<i>Hemisphere. If the vertical axis is the axis of symmetry, it will make a hemisphere, resembling a the top of a muffin.</i>



Direct the students to create each shape using sticky notes. The teacher can also use cardboard or other thick board to cut out shapes and demonstrate the rotations using a drill.

9.4.2 Activity: Rotating Pencils (continued)

7. Discuss with your partner whether the three-dimensional shape would be different if the pencil were held horizontally instead of vertically. Summarize your discussion.

Answers will vary. The figures formed by rotation could remain the same if the axis of rotation is in the same location relative to the 2D object. However, if the 2D shape keeps its orientation and the axis of rotation (pencil) is horizontal, then the 3D shape that is formed will change. The rectangle makes a cylinder with different dimensions. The triangle will vary depending on the orientation of the triangle and location of the axis. The semicircle makes a sphere.



Students may describe different three-dimensional objects depending on the type of triangle they rotated. Also, where the figure was taped to the pencil will influence the object formed. Ask probing questions about these differences while students share out their findings.

8. Draw any four-sided, two-dimensional shape on a 3x5 card. Make it as unique as possible. Cut out your shape. Draw the outline of your shape below.

Answers will vary.

9. Sketch what you think the three-dimensional figure of your four-sided two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis
<i>Answers will vary.</i>	<i>Answers will vary.</i>



The teacher may also use a digital approach on questions 8 and 9 if necessary.

10. Review your partner's shape. Then, complete the following.

- Draw your partner's shape on the grid below.
- Draw a horizontal axis of rotation.
- Draw the reflection of your partner's shape.



Answers will vary.

9.4.2 Activity: Rotating Pencils (continued)

11. Discuss with your partner how drawing a reflection could help envision the three-dimensional figure. Summarize your discussion.

Answers will vary. A reflection on the other side of the axis will help to see what the direction the shape is going as well as how the shape is folding around the axis asked.

12. Sketch what you think the three-dimensional figure of your partner's four-sided, two-dimensional shape will look like if rotated around each axis.

Horizontal Axis	Vertical Axis
<i>Answers will vary.</i>	<i>Answers will vary.</i>

13. Compare your sketches with your partner. Discuss any differences and make any adjustments.

Answers will vary.



Allow students the opportunity to notice and wonder when working with partners on question 10.



Questions 12-13 give students the opportunity to look at multiple representations given the horizontal and vertical axes.

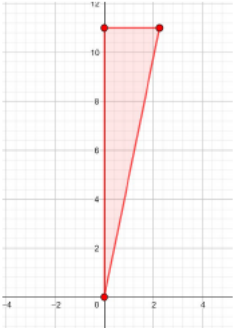
9.4.3 Guided Instruction: Rotations With Dimensions

Component Overview: Students will sketch two-dimensional and three-dimensional figures and must work with both representations. In question 1, an image of a three-dimensional object is given, and students must sketch the two-dimensional figure that would be formed. Question 2 requires students to sketch a three-dimensional object when given a two-dimensional figure and an axis of rotation.



9.4.3 Guided Instruction: Rotations with Dimensions

1. Draw a two-dimensional figure and an axis of rotation that would result in the real-world object when the two-dimensional figure is rotated around the axis of rotation.

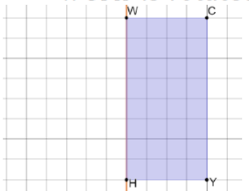
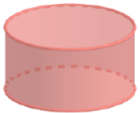
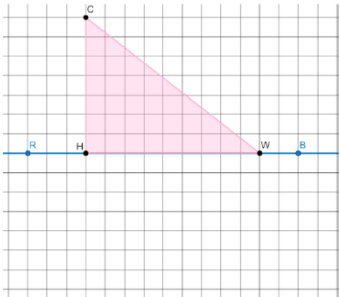
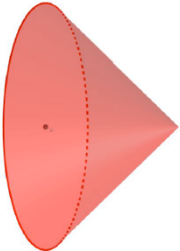
Image	Two-Dimensional Figure
A set of revolving doors with a radius of 3 feet. 	
An ice cream cone with a diameter of 4.5 centimeters (cm) and a height of 11 cm. 	



Have students who are struggling look at the pictures to help determine what two-dimensional figure is formed. Engage students in conversation about what the various cross sections of the object would look like at various locations and what part of the cross section is needed to create the object. Monitor to see if students are graphing correctly.

9.4.3 Guided Instruction: Rotations With Dimensions (continued)

2. Sketch the three-dimensional shape that is created.
Include any dimensions in units.

Two-Dimensional Shape	Three-Dimensional Shape
$WCYH$ is rotated around $\overline{WH}$. 	 Cylinder Radius: 4 units Height: 8 units
$\triangle CWH$ is rotated around $\overline{RB}$. 	 Cone Radius: 7 units Height: 9 units



How can students find the dimensions of the three-dimensional objects using the two-dimensional figures?



What are the similarities and differences between rotating the triangle around $\overline{RB}$ and rotating it around $\overline{HC}$? (Similarities are that they both create a cone shape; differences are the height and radius of the two cones will be different.)

9.4.4 Guided Practice: Rotations

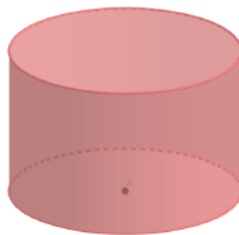
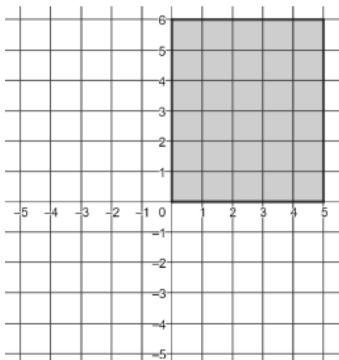
Component Overview: Students complete two questions where students sketch a three-dimensional object when given a two-dimensional figure on the coordinate plane.



9.4.4 Guided Practice: Rotations

1. Describe and sketch the three-dimensional shape that is created when the rectangle is rotated around the y -axis. Include any known dimensions.

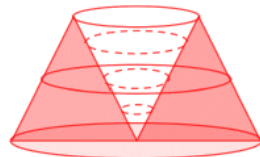
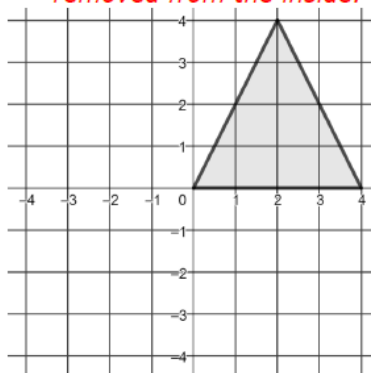
Cylinder with radius of 5 units and a height of 6 units.



For students who struggle with drawing, consider making a “bank” of figures for the students to choose from that represents what the rotated image will look like. Pay closer attention to the students’ descriptions than to their abilities to represent the figure through a drawing.

2. Describe and sketch the three-dimensional shape that is created when the isosceles triangle is rotated around the y -axis. Include any known dimensions.

Answers will vary. Truncated cone of height 4 and a radius of 4 with a cone of height 4 and radius of 2 removed from the inside.



For students who finish early, challenge them to create a third two-dimensional figure on the coordinate plane and draw the resulting three-dimensional object that is created when rotated around an axis.

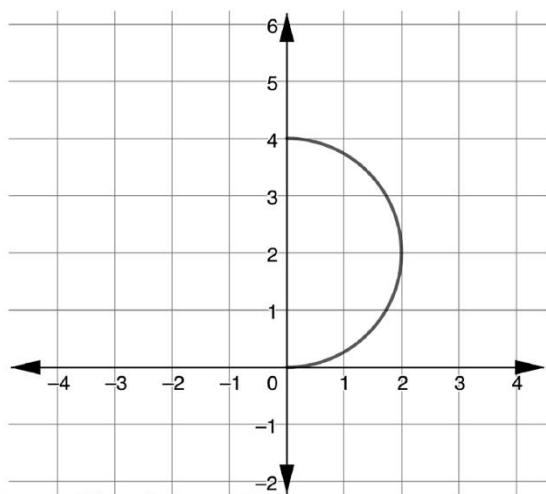
9.4.5 Wrap-Up: Ticket Out the Door

Component Overview: Students complete a question to measure their understanding of the content of the lesson.

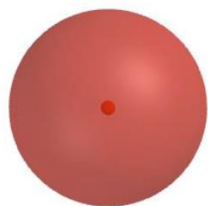


9.4.5 Wrap-Up: Ticket Out the Door

1. Describe and sketch the three-dimensional shape that is created when the semicircle is rotated around its diameter. Include any known dimensions.



Sphere with radius 2 units.



Consider using student responses from this activity for differentiation in future lessons.